11-24 第2篇

阅读理解（说明文） Why Does Popcorn Pop?

① In our school canteen, there is one small miracle every Friday afternoon. The cook pours a cup of dry yellow kernels into a wide iron pan with a little oil, puts a clear glass lid on top, and turns the heat on. For the first minute, almost nothing happens. Then — **pop! pop-pop-pop-pop! POP-POP!!** — the quiet kernels explode into white, soft, cloud-shaped pieces that fill the pan to its brim. Students stand near the window and clap without thinking. How can a hard, silent, pale-yellow seed turn, in sixty seconds, into a crispy white shape about forty times its own size? The answer lives in **four quiet rules of physics and biology** that come together in a surprising little show.

② The first rule is about **water hidden inside the seed**. A popcorn kernel is not a dry stone as it looks. Inside, under a smooth, hard outer layer called the **pericarp**, sits a small amount of **starch** — and, trapped in the middle of that starch, about **13% to 14.5% of water**. This is a very particular number. Corn grown too dry (below 12% water) refuses to pop at all; corn grown too wet (over 16%) pops weakly into small, soft "half-pops." Good popcorn kernels are **carefully dried** at farms to keep the water level exactly in this small "magic window."

③ The second rule is about **a very strong wall**. The pericarp of a popcorn kernel is unusually **tough** — much tougher than the skin of ordinary sweet corn or field corn. When the kernel is heated, the water inside turns slowly into **steam**. Because the pericarp does not easily break, the steam cannot escape. **Pressure builds up** inside the tiny seed. By the time the pan is hot enough (about 180 °C / 356 °F), the pressure inside a single kernel has reached about **9 atmospheres** — roughly the pressure inside a racing bicycle tyre. All this, inside a seed the size of a small fingernail.

④ The third rule is about **a sudden fall in pressure**. When the hot pericarp finally cannot hold any longer, it **cracks open in a fraction of a second**. The high-pressure steam bursts out, and the soft, almost-liquid hot starch inside is suddenly exposed to normal air pressure. Like a small bottle of sparkling water opened too quickly, the starch **foams outward**, trapping bubbles of steam in a soft, fast-growing network. In about one-tenth of a second, that tiny white foam **freezes into shape** as it cools — turning a hard seed, forty times larger and a million times fluffier, into the piece you will eat.

⑤ The fourth rule is a small but important one: **starch is the only kind of seed filling that does this nicely**. If you tried the same experiment with peas or rice, nothing interesting would happen — their internal walls cannot hold enough pressure, or they break long before the foam stage. **Only a narrow kind of corn, named "Zea mays**

everta," has the right mix of hard pericarp, thick starch, and the correct water content. Sweet corn (which you eat in summer) and field corn (used for animal feed or flour) have similar insides but **weaker pericarps**; their kernels can only burn, not pop. Farmers have, over thousands of years, quietly selected "everta" kernels for exactly this one small trick.

⑥ You can verify the rules in your own kitchen with a careful experiment (ask an adult first). **Experiment 1**: cut open one raw kernel — you will see the yellow starch and a surprisingly dense, smooth outer shell. **Experiment 2**: place two kernels on a plate in direct sunlight for one day to dry them out more; try to pop them the next day. Many will fail to pop, because the water has been partly lost. **Experiment 3**: compare a bag of fresh popcorn to one that is six months old in a kitchen cupboard with the lid half-open. The older bag often pops more poorly — again because of lost water. Every failed popping, in other words, is a small proof that **water is the real actor**.

⑦ A few more surprising details. First, there are **two main shapes** of popped kernel: *"mushroom"* — round and smooth, often used in caramel popcorn because they do not break easily — and *"butterfly"* — bigger, with delicate wings, preferred for cinema popcorn because they catch more salt and oil. Both come from the same kind of kernel; which one forms depends on **how quickly the kernel cools**. Second, the famous **"pop"** sound is not the sound of the pericarp breaking; it is the sound of **steam rushing out and crossing the speed of an air wave** inside the pan — a tiny sonic boom no bigger than a dot.

⑧ So the next time you watch the Friday pan in the canteen and listen to the soft, quick little explosions, you may feel you are watching a **miniature physics lesson wearing a snack costume**. A hard pericarp, a careful 13—14.5% of water, a slow climb to 9 atmospheres, and a one-tenth-second foam — each popcorn is a tiny steam engine that ends, not in a train, but in a soft white cloud you can eat. The miracle is not only that it works. **The miracle is that it has been working, in kitchens and cinemas and small school canteens, exactly the same way, for at least 5,000 years** — since the first farmers in Mexico quietly noticed what this one unusual kernel could do.

注释： kernel 颗粒；pericarp 果皮；starch 淀粉；pressure 压力；foam 泡沫

根据短文内容，选择最佳答案。

1. Why must a popcorn kernel contain the right amount of water (about 13%—14.5%) to pop well?

A. Because water tastes better than oil.
B. Because corn grown too dry (below 12% water) refuses to pop at all, and corn grown too wet (over 16%) only pops weakly into small soft "half-pops"; farmers carefully dry kernels to keep the water level inside a small "magic window."

C. Because wet kernels are healthier.
D. Because dry kernels are cheaper.

2. What makes the pressure inside a heated kernel rise to about 9 atmospheres?

A. The starch pressing on the pericarp.
B. The pericarp is unusually tough and does not easily break, so the water inside turns slowly into steam but cannot escape — pressure builds up until it reaches about 9 atmospheres, roughly the pressure inside a racing bicycle tyre.
C. The oil in the pan.
D. The noise of the kernel.

3. According to the passage, what happens in the fraction of a second after the pericarp cracks?

A. The kernel burns.
B. The high-pressure steam bursts out; the soft hot starch, suddenly under normal pressure, foams outward like newly opened sparkling water, trapping bubbles of steam; within about one-tenth of a second this foam cools and freezes into the fluffy white shape we eat.
C. The kernel becomes smaller.
D. The kernel turns into sugar.

4. Why can ordinary sweet corn or field corn NOT pop in the same way?

A. Because they are not yellow.
B. Because they have weaker pericarps; their kernels can only burn, not build up enough pressure to pop — only a narrow kind of corn named "Zea mays everta" has the right mix of hard pericarp, thick starch, and correct water content.
C. Because they have no starch.
D. Because they are forbidden.

5. What is the writer's main point in the last paragraph?

A. Popcorn is only a snack and nothing more.
B. Each popcorn is like a tiny steam engine — a small, careful piece of physics in a snack costume; the real miracle is that popcorn has been popping the same way for at least 5,000 years, since the first farmers in Mexico noticed what this one unusual kernel could do.
C. Popcorn should replace other snacks.
D. Popcorn is dangerous.

答案与解析

1.
【答案】B
【题型】细节理解题
【定位】第②段"magic window"
【解析】13%—14.5% 水分直引。A、C、D 均错。故选 B。

2.
【答案】B
【题型】细节理解题
【定位】第③段压力积累
【解析】9 atm 与 racing tyre 类比直引。A、C、D 均错。故选 B。

3.
【答案】B
【题型】细节理解题
【定位】第④段爆裂瞬间
【解析】泡沫 + 1/10 秒冷却直引。A、C、D 均错。故选 B。

4.
【答案】B
【题型】细节理解题
【定位】第⑤段品种对比
【解析】"everta"与其他玉米对比直引。A、C、D 均错。故选 B。

5.
【答案】B
【题型】推理判断题
【定位】第⑧段结尾
【解析】"5000 年"微型蒸汽机主旨。A、C、D 均偏离。故选 B。